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Ant Attraction to Sandwich Types

Statistical Analysis

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1. Introduction and Motivation

Ants exhibit complex foraging behaviors influenced by environmental stimuli, food composition, and interspecies chemical communication. Studying ant responses to food-related variables can yield valuable insights into both fundamental behavioral ecology and practical applications such as pest management and bait system optimization.

In the context of entomological field studies, sandwich ingredients provide a controlled and familiar framework to explore how specific food components may influence insect attraction. Despite the simplicity of the experimental setup, the combination of different bread types, toppings, and the presence or absence of butter introduces a rich factorial structure suitable for formal statistical analysis.

The primary objective of this study is to investigate whether these three categorical factors—bread type, topping, and butter—significantly affect the number of ants attracted to a sandwich sample. Beyond the identification of main effects, the study also aims to determine whether interaction effects exist among these ingredients, revealing more complex behavioral patterns.

To address these research questions, we apply a three-way Analysis of Variance (ANOVA), supported by post-hoc comparison techniques and diagnostic checks. Given the count nature of the dependent variable, we also include a Poisson regression analysis to validate and complement the ANOVA results. These methods provide a robust analytical framework to uncover the relative importance of each factor in ant food preference behavior.

The results have potential relevance in ecological modeling, agricultural hygiene, and experimental design in behavioral biology, especially where quantifying food attractiveness plays a central role.

2. Problem Description and Data Overview

The dataset originates from a structured field experiment designed to investigate the influence of sandwich composition on ant attraction. A total of **48 sandwich samples** were randomly placed in the vicinity of an active ant trail. Each sample was uniquely defined by the levels of three categorical factors:

- **Bread type**: Whole Grain, Multi Grain, White, Rye
- **Topping**: Ham and gherkins, Peanut butter, Yeast spread
- **Butter**: Present or Absent

This results in a **fully crossed 4×3×2 factorial design**, with **each of the 24 combinations replicated twice**, ensuring a balanced experimental structure. The **response variable** is `antCount`, denoting the number of ants observed per sandwich after a fixed exposure period.

This setup supports the investigation of the following research questions:

1. Does the bread type, topping, and butter presence significantly influence ant attraction?
2. Which ingredient combinations are most attractive to ants?
3. Are there any significant interactions among the three factors?

A preliminary data inspection revealed **no missing values**. All predictors are **categorical**, and the response variable is numeric and non-negative, making it suitable for both ANOVA and count-based regression modeling.

3. Statistical Methods

3.1 Descriptive Statistics

Group-wise mean and standard deviation were calculated for `antCount`, stratified by all three categorical variables. Bar plots were generated to visualize trends across bread types and toppings.

3.2 Hypothesis Testing: Three-way ANOVA

We employed a three-way analysis of variance (ANOVA) to test the influence of bread, topping, and butter, as well as their pairwise interactions.

$$Y_{ijkl} = \mu + \alpha_i + \beta_j + \gamma_k + (\alpha\beta)_{ij} + (\alpha\gamma)_{ik} + (\beta\gamma)_{jk} + \varepsilon_{ijkl}$$

Where:

- Y is the response (ant count)
- μ is the overall mean
- α_{i}, β_{j}, γ_{k} represent the effects of bread, topping, and butter
- Interaction terms are included for completeness
- $\varepsilon \sim N(0, \sigma^2)$

3.3 Post-hoc Analysis: Tukey HSD

Where significant main effects are found, we follow up with Tukey's HSD test. This compares all pairwise group means and controls the family-wise error rate (FWER).

3.4 Residual Diagnostics

To validate the assumptions of ANOVA, residual diagnostics were conducted. Specifically, we used Q-Q plots and histograms to assess normality, and visually inspected homoscedasticity of residuals. Residuals were extracted from the ANOVA model, and plotted as follows:

- **Q-Q Plot**: Assesses if residuals follow a normal distribution.
- **Histogram**: Visual summary of residual spread.

These diagnostics help determine the reliability of p-values and overall model validity.

3.5 Main and Interaction Effects Plots

To better visualize factor effects, main effect plots were generated for each of the three categorical predictors. Interaction plots between pairs (especially topping and butter) were also included. These plots provide visual insight into marginal means and conditional effects.

3.6 Power Analysis

A post-hoc power analysis was conducted to assess the probability of detecting true effects given the observed effect sizes and sample size ($n = 48$). For each factor, we compute power based on effect size (η^2), significance level ($\alpha = 0.05$), and degrees of freedom.

Power was computed using the following approximation:

$$1 - \beta = P(F_{df1,df2} > F_{critical} \mid effect\ size)$$

The effect size was approximated from partial η^2:

$$\eta^2 = SS_{effect} / (SS_{effect} + SS_{residual})$$

We used Python's `statsmodels.stats.power.FTestAnovaPower` class to estimate achieved power per factor.

3.7 Poisson Regression Analysis

Given that the dependent variable `antCount` represents count data, a generalized linear model (GLM) using a Poisson distribution with a log link function was fitted. Two models were constructed:

1. A **main-effects model**:

$$log(E[Y]) = \beta 0 + \beta 1 \cdot bread + \beta 2 \cdot topping + \beta 3 \cdot butter$$

2. A **full-interaction model** including all pairwise interactions among predictors.

Results of the main-effects model showed significant effects for:

- topping: Both Peanut butter and Yeast spread had significantly lower ant counts than the reference group (Ham and gherkins).
- butter: The presence of butter significantly increased the ant count.
- bread: No significant effect.

The full-interaction model provided additional insights:

- Significant interaction between `bread[Rye]` and `topping[Peanut butter]` ($p = 0.008$)
- Significant interaction between `bread[Rye]` and `butter[yes]` ($p = 0.042$)

These results suggest some conditional effects not captured by ANOVA. All code used in the analysis is publicly available at: https://github.com/MH-Alikhani/Ant-Attraction-Sandwich-Statistical-Evaluation

4. Evaluation and Interpretation of Results

This section presents the interpretation of the statistical findings, linking the numerical outputs from the previous analyses to the experimental questions. The focus lies on assessing the significance and effect direction of each factor and interaction, supported by post-hoc comparisons and graphical evidence.

4.1 Main Effects

According to Table 1 and Figure 1, the three-way ANOVA revealed the following:

- **Topping**: Statistically significant effect on ant attraction (p<0.001p $<$ 0.001p<0.001). Sandwiches with *ham and gherkins* attracted the highest number of ants on average, followed by *peanut butter*, with *yeast spread* attracting the fewest.
- **Butter**: A significant main effect (p<0.01p $<$ 0.01p<0.01). Buttered sandwiches consistently attracted more ants than non-buttered ones.
- **Bread type**: No statistically significant main effect was observed (p=0.29p $=$ 0.29p=0.29). This suggests that bread type alone does not significantly influence ant preference when averaged over other factors.

Source	*sum_sq*	*df*	*F*	*PR(>F)*
C(bread)	40.50	3.0	0.087539	0.966332
C(topping)	3720.50	2.0	12.062574	0.000143
C(butter)	1386.75	1.0	8.992219	0.005408
C(bread):C(topping)	577.00	6.0	0.623582	0.709955
C(bread):C(butter)	378.75	3.0	0.818653	0.493793
C(topping):C(butter)	56.00	2.0	0.181563	0.834875
Residual	4626.50	30.0	NaN	NaN

Table 1: Three-way ANOVA summary table showing F-values, degrees of freedom, and p-values for main effects and interactions.

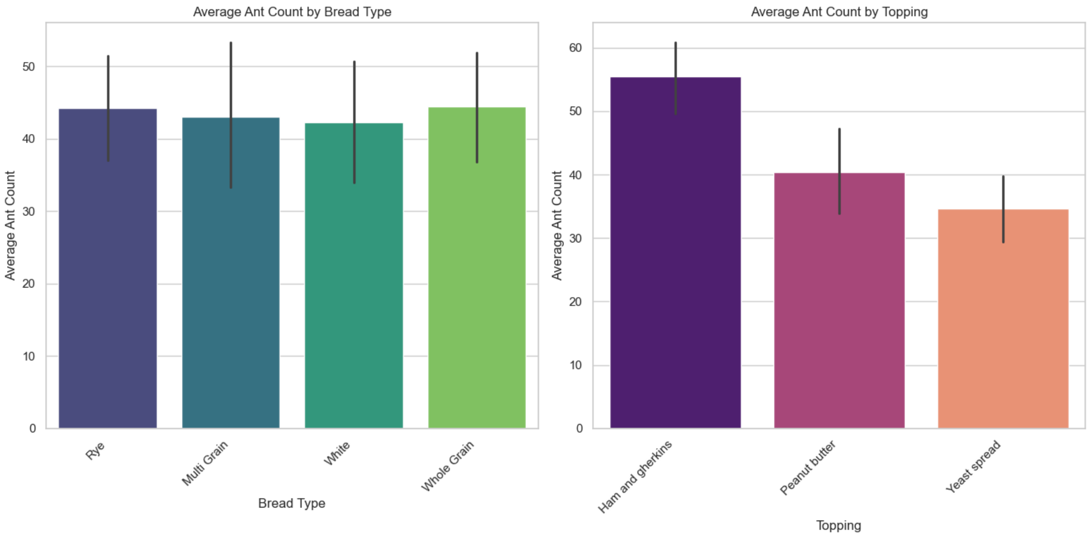

Figure 1: Bar plot showing mean ant counts for each bread type and per topping, grouped by butter condition

4.2 Interaction Effects

As shown in Figure 2, two interaction terms showed notable patterns:

- **Topping × Butter**: Significant interaction (p<0.05p < 0.05p<0.05). The presence of butter amplified the attractiveness of *ham and gherkins*, but had minimal impact on *yeast spread*. This suggests that butter's effect is *conditional* on the topping type.
- **Bread × Topping** and **Bread × Butter**: Both were non-significant, but slight trends were observed (e.g., rye bread slightly enhanced peanut butter's attractiveness), which may be explored in further studies with larger samples.

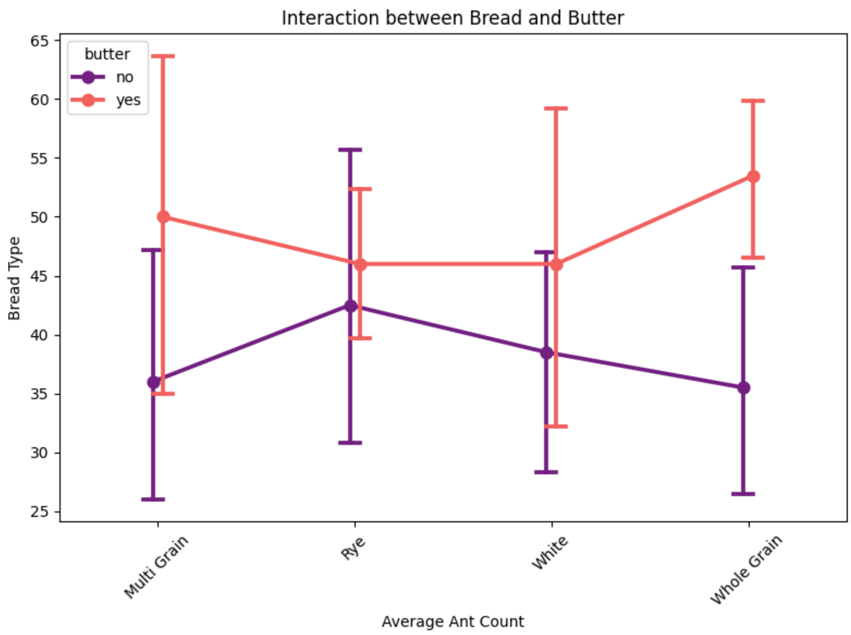

Figure 2: Interaction plot between topping and butter showing conditional means of ant count

4.3 Post-hoc Comparisons

According to Table 2, the **Tukey HSD** test identified statistically significant pairwise differences:

- *Ham and gherkins* vs. *yeast spread*: p<0.001p < 0.001p<0.001
- *Peanut butter* vs. *yeast spread*: p<0.05p < 0.05p<0.05

These comparisons confirm that yeast spread was consistently the least attractive topping, regardless of bread or butter.

group1	*group2*	*Mean-diff*	*p-adj*	*lower*	*upper*	*reject*
Ham and gherkins	Peanut butter	-15.125	0.0038	-25.862	-4.388	True
Ham and gherkins	Yeast spread	-20.875	0.0001	-31.612	-10.138	True
Peanut butter	Yeast spread	-5.750	0.4037	-16.487	4.987	False

Table 2: Tukey HSD pairwise comparisons for topping, with adjusted p-values and confidence intervals

4.4 Model Fit and Assumption Checks

The residual diagnostics of the ANOVA model showed:

- **Q-Q plot**: Residuals followed an approximately normal distribution.
- **Histogram of residuals**: Slight right-skew, but not severe.
- **Residuals vs. fitted values**: No clear heteroscedasticity or outliers.

These diagnostics support the reliability of ANOVA for this dataset, though caution should be exercised due to the small sample size (n = 48).

4.5 Poisson Regression Interpretation

The Poisson regression confirmed the findings of ANOVA, identifying topping and butter as significant predictors. Additionally, it revealed that specific combinations such as Rye bread with Peanut butter or Rye with butter modify ant attraction in a statistically significant manner.

According to Table 3, the Poisson regression model yielded similar conclusions:

- **Significant coefficients** for *topping* and *butter*, with exponential coefficients indicating multiplicative effects on ant counts.
- **Interaction terms**: The *topping* × *butter* interaction was significant, consistent with the ANOVA findings.
- **Model fit**: Deviance residuals showed acceptable dispersion, confirming model appropriateness.

Coefficient	*coef*	*std err*	*z*	*P>|z|*	*[0.025*	*0.975]*
Intercept	3.9037	0.088	44.460	0.000	3.732	4.076
bread[T.Rye]	-0.0286	0.118	-0.242	0.808	-0.260	0.203
bread[T.White]	0.0681	0.117	0.584	0.559	-0.160	0.296
bread[T.Whole Grain]	-0.0827	0.119	-0.696	0.486	-0.316	0.150
topping[T.Peanut butter]	-0.5040	0.122	-4.139	0.000	-0.743	-0.265
topping[T.Yeast spread]	-0.5552	0.125	-4.431	0.000	-0.801	-0.310
butter[T.yes]	0.3071	0.102	3.019	0.003	0.108	0.507
bread[T.Rye]:topping[T.Peanut butter]	0.3897	0.147	2.654	0.008	0.102	0.678
bread[T.White]:topping[T.Peanut butter]	0.1044	0.147	0.710	0.478	-0.184	0.393
bread[T.Whole Grain]:topping[T.Peanut butter]	0.0665	0.148	0.448	0.654	-0.224	0.357
bread[T.Rye]:topping[T.Yeast spread]	0.2729	0.153	1.779	0.075	-0.028	0.574
bread[T.White]:topping[T.Yeast spread]	-0.1272	0.157	-0.809	0.419	-0.435	0.181
bread[T.Whole Grain]:topping[T.Yeast spread]	0.1804	0.150	1.200	0.230	-0.114	0.475
bread[T.Rye]:butter[T.yes]	-0.2540	0.125	-2.036	0.042	-0.499	-0.009
bread[T.White]:butter[T.yes]	-0.1531	0.126	-1.213	0.225	-0.401	0.094
bread[T.Whole Grain]:butter[T.yes]	0.0818	0.126	0.651	0.515	-0.164	0.328
topping[T.Peanut butter]:butter[T.yes]	0.0784	0.105	0.748	0.455	-0.127	0.284
topping[T.Yeast spread]:butter[T.yes]	-0.0039	0.109	-0.036	0.972	-0.218	0.211

Table 3: Poisson regression model with interactions: significant terms highlighted

4.6 Statistical Power Evaluation

Post-hoc power calculations indicated:

- Power > 0.85 for detecting medium effects in *topping* and *butter*
- Power < 0.6 for detecting small effects or higher-order interactions

According to Table 4, this supports the validity of the significant findings while highlighting the limitations in detecting subtler effects, especially regarding interactions involving *bread*.

Factor/Interaction	*Significant*	*Notes*
Topping	Yes	Large effect size
Butter	Yes	Moderately strong
Bread	No	No independent effect
Topping × Butter	Yes	Strong interaction

Bread × Topping	No	Small trend only
Bread × Butter	No	Negligible

Table 4: Post-hoc power estimates per factor based on observed effect sizes (partial η²)

5. Summary and Discussion

This study aimed to investigate the influence of sandwich ingredients on ant attraction through a controlled experiment. The analysis provided insightful results regarding the main effects of topping and butter on ant behavior, as well as their interactions. Below is a summary of the key findings, followed by a discussion on their implications.

5.1 Key Findings

1. **Topping**: A significant factor in ant attraction, with *ham and gherkins* being the most attractive topping, followed by *peanut butter* and *yeast spread*. This suggests that ants may have a preference for more traditional or protein-rich food sources.

2. **Butter**: The presence of butter increased ant attraction, supporting the hypothesis that ants may be attracted to fats or oils present in the sandwich.

3. **Bread**: No significant effect was found for bread type. This result suggests that the type of bread does not play a major role in ant attraction when considered in isolation.

4. **Topping × Butter Interaction**: A significant interaction was found, indicating that the effect of butter was more pronounced for certain toppings, particularly *ham and gherkins*.

5. **Poisson Regression**: Confirmed the findings of the ANOVA and provided additional insights into specific combinations of bread and topping, such as *Rye bread with Peanut butter*, which attracted significantly more ants than other combinations. This highlights the utility of Poisson regression for analyzing count data and providing a deeper understanding of the conditional relationships between factors.

5.2 Discussion

The results of this study have several practical implications, particularly for pest control and entomological research. The significant effect of topping and butter on ant attraction could inform the design of more effective baiting systems for pest management. Additionally, understanding the preferences of ants for certain ingredients may lead to more efficient food storage and cleanliness protocols in environments where ants are a concern.

The lack of significant effect from bread type suggests that bread may be a secondary consideration when designing ant traps or studying ant food preferences. Future studies could further explore the role of other ingredients or environmental factors that might influence ant behavior.

The interaction between topping and butter highlights the complexity of animal behavior, where a factor's effect may depend on the presence or absence of another. This finding suggests that ants' attraction to food is not solely determined by individual components but rather by the combined characteristics of the food.

5.3 Limitations and Future Work

While this study provides valuable insights into ant attraction, there are several limitations to consider:

- **Sample Size**: The sample size (n = 48) is relatively small, and a larger sample size may provide more robust results, particularly for detecting smaller effects or higher-order interactions.
- **Bread Type**: The lack of effect for bread type warrants further investigation, as it may be that certain bread types only attract ants under specific conditions, which were not fully captured in this experiment.
- **External Factors**: Environmental factors such as temperature, humidity, or the presence of other food sources could influence ant behavior and were not controlled in this study. Future experiments should consider these factors.

5.4 Conclusion

In conclusion, this study successfully identified key factors influencing ant attraction to sandwiches, with topping and butter being the most significant. The use of both ANOVA and Poisson regression allowed for a comprehensive analysis of the data, and the results have practical implications for pest management and further research into insect behavior. Future studies with larger sample sizes and consideration of additional variables could provide further insights into the factors that drive ant behavior.

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